TAVE

Wearable Computer Control Without a Mouse

Natural hand movement. Simple muscle gestures. Full cursor control.

FORM FACTOR
Wrist/Forearm Band

TEAM
TAVE Engineering Team



Motion Sensor

Muscle Sensor

Speaker Notes: *Welcome judges and panel. Today we are introducing TAVE, a wearable wristband system designed to provide precise computer cursor and click control using standard physiological sensing without requiring a desktop surface.*

THE PROBLEM

TAVE

Computer Control Is Still Tied to a Surface

Desktop Mouse Dependency

Traditional mice require a continuous flat physical surface to operate.

Presenter Remote & Air Mouse Limits

Presenter remotes offer limited controls; air mice lack steady pointing accuracy.

Accessibility Barriers

Users with limited hand mobility struggle with rigid traditional hardware.

CORE QUESTION

Can we control a computer naturally without holding a mouse or using a desk?

Speaker Notes: *Traditional computer input hasn't evolved beyond the desk desk constraint. Presenter remotes lack precision, air mice suffer from shaking, and accessibility remains an issue. Our core challenge was freeing interaction from flat surfaces entirely.*

OUR IDEA

A Mouse You Wear

Hand Movement
Natural, fluid spatial tracking

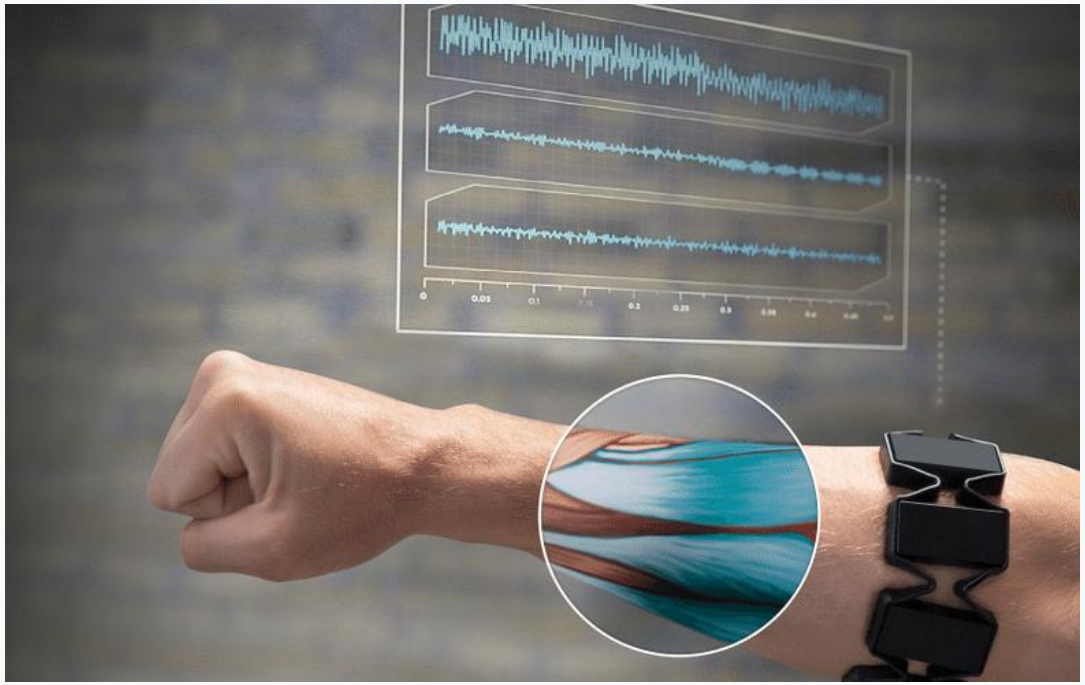
Move your wrist → Move the cursor

Muscle Gesture
Subtle forearm muscle activation

Make a small intentional squeeze → Click

Wireless Connection
Connects automatically as a standard mouse

Bluetooth HID Protocol



Move → Point

Squeeze → Click

Speaker Notes: *TAVE translates natural wrist motion into smooth pointer trajectory, paired with physiological muscle squeeze sensing for instant left-clicks. It pairs via standard Bluetooth without requiring custom client software.*

SYSTEM ARCHITECTURE

TAVE

Two Sensors, Two Jobs

1. MOTION SENSOR

Detects wrist movement

Purpose: Cursor Position

2. MUSCLE SENSOR

Detects electrical muscle activity

Purpose: Click Detection

HARDWARE SIGNAL PROCESSING FLOW



Speaker Notes: *By separating continuous spatial positioning from discrete click triggers using dedicated sensors, TAVE eliminates false triggers. Embedded firmware processes raw data onboard before transmitting over standard Bluetooth HID.*

THE MAIN PROBLEM

TAVE

Involuntary Cursor Movement During Clicking

Physical Strain & Drift

When a user performs a click, muscle contraction causes a small involuntary hand movement.

- ✕ **Cursor Displacements:** Cursor moves away from intended target before click registers.
- ✕ **Missed Interactions:** Users intend to click one point, but land elsewhere.
- ⚠ **Air Mouse Flaw:** Primary flaw identified in existing gesture controllers.

INVOLUNTARY DRIFT DIAGRAM



Result: Inaccurate input execution

Speaker Notes: *Air mice traditionally fail because muscular tightening during clicking causes involuntary hand twitching. The cursor shifts right as the trigger fires, missing small buttons and links.*

HERO ENGINEERING FEATURE

PATENT-PENDING ALGORITHM

TAVE

Click Stabilization

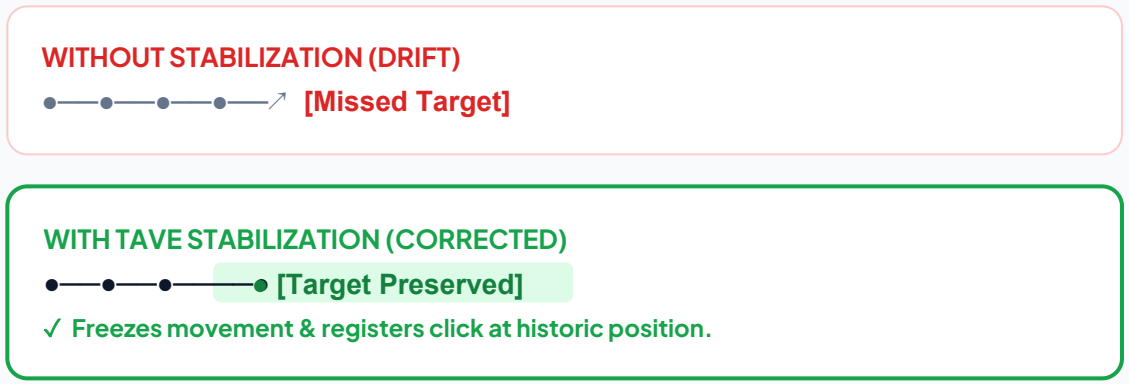
ALGORITHM EXECUTION PIPELINE

1. **Detect:** Muscle sensor triggers intent threshold.
2. **Freeze:** Pause live motion sensor trajectory.
3. **Rollback:** Revert to cursor position prior to muscle squeeze.
4. **Click:** Execute click at preserved intended target.

Detect → Stabilize → Click

Goal: Completely eliminate twitch movement from the click phase.

BEFORE VS AFTER STABILIZATION



Speaker Notes: *This is our hero technical innovation. TAVE buffers recent cursor positions continuously. When the muscle sensor detects a squeeze, firmware rolls back cursor positioning by milliseconds to register the click exactly where you intended.*

RELIABILITY & ACCURACY

TAVE

Preventing Accidental Clicks

Normal Muscle Activity

→ No Click Triggered

Filters ambient arm tension and movement

Intentional Muscle Squeeze

→ Click Registered

Recognizes sharp deliberate firing

QUICK CALIBRATION WORKFLOW

1. Relax
Establishes baseline baseline noise

→

2. Normal Squeeze
Samples intentional click intensity

→

3. System Learns
Sets adaptive thresholding

→

4. Ready
Tailored to user physiology

Speaker Notes: *Everyone's arm muscle signals are unique. TAVE incorporates a quick 5-second calibration procedure that samples background muscle resting state and deliberate squeeze strength to adapt sensitivity perfectly.*

MOTION SIGNAL PROCESSING

TAVE

Smooth Cursor Control

Adaptive Filtering Engine

Eliminates small micro-tremors without introducing noticeable motion latency.

Slow Movement

High Smoothing: Maximizes precision for small desktop UI targets.

Fast Movement

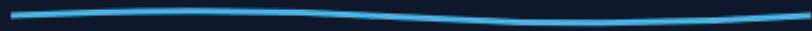
Low Smoothing: Preserves instant high-speed responsiveness.

SIGNAL SMOOTHING GRAPHIC

RAW MOTION SIGNAL (UNFILTERED SHAKE)



TAVE FILTERED SIGNAL (ADAPTIVE SMOOTH)



Result: Clean pointer trajectories with zero latency penalty.

Speaker Notes: *Raw sensor streams capture physiological arm shaking. We use an adaptive low-pass filter: when you move slowly, smoothing increases for pinpoint target selection; when you move quickly, smoothing drops for zero lag.*

Competitor Landscape

Existing Solutions

Wearable and motion-based computer controllers already exist.

Product	Technology / Approach	Main Use	Price
Mudra Link	Neural sensing + wrist control	General device control	$249 promo / $299 regular
TapXR	Finger gestures + motion sensing	Air mouse + keyboard	$199
Logitech Spotlight	Motion-based remote	Presentations	$119.99
WowMouse	Smartwatch-based gesture control	Wrist-based computer control	$4.99 app + smartwatch
TAVE	Motion + muscle sensing	Wearable computer control	Target: affordable

Market Insight

The market already proves that people are interested in surface-free computer interaction.
Our goal is not to claim that wearable computer control is new.
Our goal is to improve the interaction itself.

Competitor Comparison

Feature	Mudra Link	TapXR	Spotlight	WowMouse	TAVE
Wearable Hardware	✓	✓	✗	✓ *	✓ Dedicated
Surface-Free Control	✓	✓	✓	✓	✓ Yes
Muscle-Based Click	—	—	✗	✗	✓ Forearm EMG
No Smartwatch Required	✓	✓	✓	✗	✓ Standalone
Click Stabilization	—	—	✗	✗	✓ Integrated
Affordable Target	✗ ($249+)	✗ ($199)	✗ ($119)	—	✓ $50–$100 Target

** WowMouse requires compatible smartwatch hardware. Note: Wearable computer control exists; TAVE's innovation focuses on interaction reliability and accessibility.*

Speaker Notes: *We are candid that gesture input isn't brand new. However, current options either cost hundreds of dollars or require third-party smartwatches. TAVE targets standard standalone control with built-in click stabilization.*

DIFFERENTIATION

TAVE

What Makes TAVE Different?

1. Muscle-Based Clicking

Uses subtle forearm muscle contraction instead of awkward finger gestures or physical button presses.

2. Click Stabilization

Specifically engineered algorithm that cancels involuntary motion drift during the exact moment of clicking.

3. No Smartwatch Required

Dedicated standalone device allows total engineering control over sensor position, firmware, and power management.

4. Simple Interaction

No complex chorded gesture memorization. Straightforward physics: Move to point, Squeeze to click.

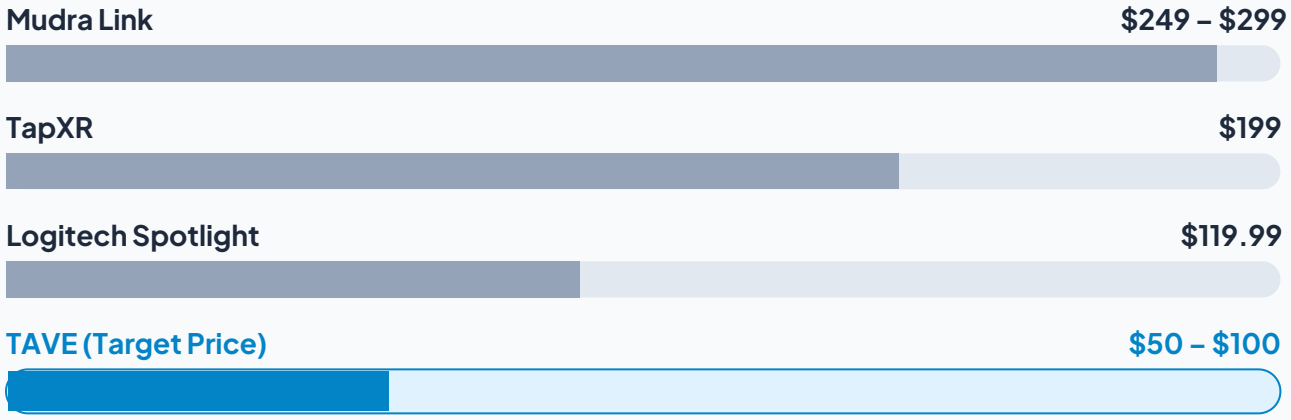
Speaker Notes: *Our defensibility comes from four integrated advantages. By controlling the dedicated hardware end-to-end, we keep interaction effortless without forcing users to wear an expensive smartwatch or learn complex chorded finger twitches.*

COMMERCIAL STRATEGY

TAVE

The Price Opportunity

MARKET PRICE COMPARISON ($ USD)



TARGET COST POSITION

$50 – $100

*Target, not final. Dependent on component sourcing, manufacturing volume, battery, and enclosure scale.

Market Insight

Existing devices prove consumer demand for surface-free input, but premium $200+ pricing restricts mass consumer adoption.

Speaker Notes: *Premium wearable input devices charge $200 to $300. We are engineering TAVE specifically for component cost efficiency, targeting a $50 to $100 retail price bracket to make surface-free control widely accessible.*

Applications & Smart Modes

TARGET USE CASES

- **Computer Control:** Standard mouse navigation.
- **Education:** Untethered presenter navigation.
- **Accessibility:** Surface-free motor assistance.
- **AR / VR:** Compact spatial UI interaction.

INTELLIGENT FIRMWARE MODES

Mouse Mode

Wrist → Cursor | Squeeze → Left Click

Presentation Mode

Wrist Flick → Slide Next/Prev | Squeeze → Select

Custom Gesture Mode

Short Squeeze = Left Click | Long Squeeze = Right Click

Speaker Notes: *Beyond general desktop computing, TAVE fits education classrooms, motor accessibility, and lightweight AR/VR navigation. Firmware supports seamless mode switching between standard mouse, presentation clicker, and custom macros.*

ENGINEERING RIGOR

Challenges & Success Metrics

MAIN ENGINEERING CHALLENGES

1. **Motion Tracking:** Stable wrist motion translation.
2. **EMG Detection:** Filtering ambient arm tension.
3. **Click Accuracy:** Eliminating displacement during clicks.
4. **Low Response Time:** Minimizing Bluetooth processing latency.
5. **User Comfort:** Ergonomic wristband design.
6. **Adaptability:** Accommodating varied user muscle profiles.
7. **Battery Life:** Optimizing power draw for all-day use.

TESTING & VALIDATION METRICS

- **Click Accuracy Rate:** % of successful target clicks.
- **Displacement Distance:** Pixel drift measured during squeeze.
- ⚡ **Response Latency:** Milliseconds from squeeze to OS click action.
- **Cursor Stability Index:** Jitter reduction under ambient noise.
- **Calibration Speed:** Time required for new user onboarding.

Speaker Notes: *We validate TAVE through measurable quantitative benchmarks. Rather than subjective demos, we test cursor jitter, click latency, and target displacement distance to guarantee commercial engineering quality.*

5-Phase Product Roadmap

PHASE 1

Basic Prototype

Breadboard hardware with motion, EMG, micro, and Bluetooth.

PHASE 2

Basic Control

Implement cursor translation and basic click detection algorithms.

PHASE 3

Accuracy & Filter

Integrate click stabilization, smoothing filters, and user calibration.

PHASE 4

User Testing

Benchmarking accuracy, ergonomics, and latency across test subjects.

PHASE 5

Product Prototype

Custom PCB, molded wristband, integrated battery, and companion app.

Speaker Notes: *Our development roadmap takes TAVE systematically from benchtop sensor validation up through firmware accuracy optimization, extensive multi-subject user testing, and final custom enclosure tooling.*

A New Way to Interact With Computers

Point & Click
Natural motion tracking with muscle squeeze execution

Move Freely
Surface-free wireless interaction for standard desktop OS

Adapt & Connect
Personalized EMG calibration with standard Bluetooth HID

TAVE: Wear it. Move naturally. Control digitally.

Speaker Notes: *We are not attempting to build another basic presentation clicker or air mouse. TAVE represents an intuitive, reliable wearable computer interface that makes digital interaction surface-free, natural, and accessible. Thank you, and we welcome your questions.*